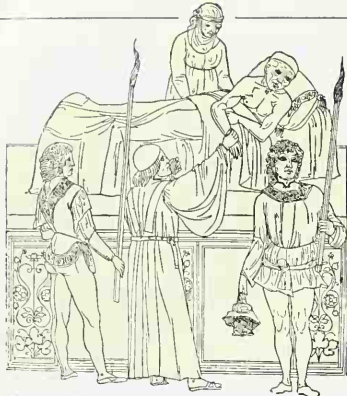


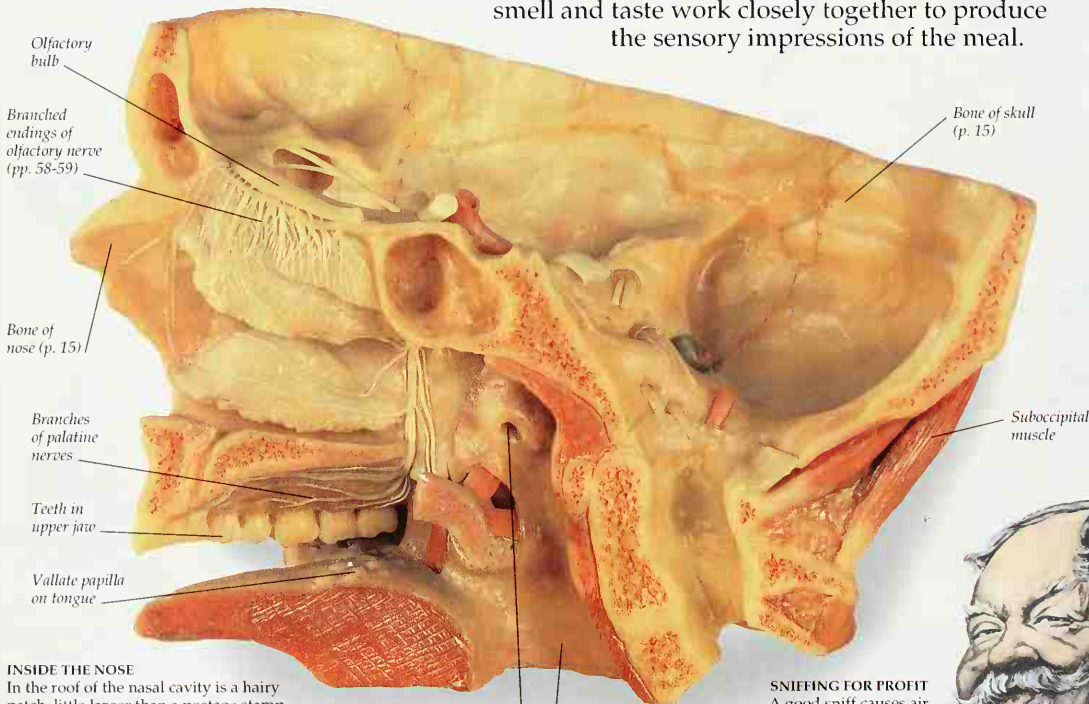


MALODOROUS AIR

For centuries, physicians suspected that diseases could spread by foul smells. For example, malaria means "bad air" in Italian. This Middle Ages physician holds a pomander – a sachet of aromatic herbs – to his nose. As well as masking the stench, the pomander was thought to protect the physician against the risk of catching the patient's ailment – in this case the Black Death.

Smell and taste

IN COMMON WITH OTHER MAMMALS, the human body has senses of taste and smell to monitor nutrients going into the digestive system. Poisonous, rotting, or unfamiliar foods and drinks produce foul smells and sharp tastes, which warn against eating or drinking. Smell is also an early-warning system for tainted air, smoke, and other dangers. These senses are often a source of pleasure, as well as warning. Smell, or olfaction, detects the scents of herbs, spices, and perfumes, and taste, or gustation, picks out the flavors of a good meal. Smell and taste are known as the chemical senses, or chemosenses. The nose and tongue detect molecules of certain chemicals, either odor molecules floating in the air or flavor molecules in food and drink. The presence of the chemicals triggers sensors on microscopic cells in the nose's roof and on the tongue's surface. The cells make nerve signals that travel to the brain for analysis by its smell and taste centers. During eating and drinking, smell and taste work closely together to produce the sensory impressions of the meal.



INSIDE THE NOSE

In the roof of the nasal cavity is a hairy patch, little larger than a postage stamp. This is the olfactory epithelium (not seen here), of thousands of cells specialized to detect odors. There may be between six and 30 different types of sensors on the cells' surfaces. When odor molecules from the air land on them, they produce nerve impulses. These are passed to the olfactory bulb, where they are sorted and pre-processed and then sent along the olfactory nerve to the brain.

Opening of eustachian tube (pp. 54-55)

Pharynx – shared passageway for food, drink, and air

SNIFFING FOR PROFIT

A good sniff causes air currents to swirl up in the nasal cavity's domed roof. This brings more odor molecules into contact with the olfactory sensors. Smell experts make their living testing scents and aromas – from flowers and perfumes, to wines and whiskies.

